NAME: Praneetha Sreedharala
SRN: PES1UG24AM281

```
C:\Users\bjpsr\OneDrive\Desktop>cd L5_42_student_grade_system

C:\Users\bjpsr\OneDrive\Desktop\L5_42_student_grade_system>dir
 Volume in drive C is Windows
 Volume Serial Number is DA1D-6DA3

 Directory of C:\Users\bjpsr\OneDrive\Desktop\L5_42_student_grade_system

27-10-2026  14:09    <DIR>          .
27-10-2026  14:09    <DIR>          ..
27-10-2026  14:09             2,024 README.md
27-10-2026  14:09             2,107 student_grade_system.py
               2 File(s)          4,131 bytes
               2 Dir(s)  245,364,142,080 bytes free

C:\Users\bjpsr\OneDrive\Desktop\L5_42_student_grade_system>
```

```
(venv) C:\Users\bjpsr\OneDrive\Desktop\L5_42_student_grade_system>python -m pylint --version
pylint 4.1.3
astroid 4.3.4
Python 3.11.0 (main, Oct 24 2022, 18:26:48) [MSC v.1933 64 bit (AMD64)]

(venv) C:\Users\bjpsr\OneDrive\Desktop\L5_42_student_grade_system>python -m bandit --version
__main__.py 1.9.4
  python version = 3.11.0 (main, Oct 24 2022, 18:26:48) [MSC v.1933 64 bit (AMD64)]

(venv) C:\Users\bjpsr\OneDrive\Desktop\L5_42_student_grade_system>python -m flake8 --version
7.4.1 (mccabe: 0.7.0, pycodestyle: 2.15.0, pyflakes: 4.0.3) CPython 3.11.0 on Windows
```

original program output

```
(venv) C:\Users\bjpsr\OneDrive\Desktop\L5_42_student_grade_system>python student_grade_system.py
Is S01 passing: True
Calculated GPA: 4.35
Student Hash: 4cfe35d8825ba802868fabe687dc860d
Registration PIN: 5033
Grade sheet exported to: grade_sheet.txt
Portal token active: ADMIN_GRADE_PORTAL_TOKEN_2026
```

Original static analysis

```
(venv) C:\Users\bjpsr\OneDrive\Desktop\L5_42_student_grade_system>python -m pylint student_grade_system.py > pylint_repo
rt_before.txt 2>&1

(venv) C:\Users\bjpsr\OneDrive\Desktop\L5_42_student_grade_system>python -m bandit -r student_grade_system.py > bandit_r
eport_before.txt 2>&1

(venv) C:\Users\bjpsr\OneDrive\Desktop\L5_42_student_grade_system>python -m flake8 student_grade_system.py > flake8_repo
rt_before.txt 2>&1

(venv) C:\Users\bjpsr\OneDrive\Desktop\L5_42_student_grade_system>dir *_report_before.txt
 Volume in drive C is Windows
 Volume Serial Number is DA1D-6DA3

 Directory of C:\Users\bjpsr\OneDrive\Desktop\L5_42_student_grade_system

27-10-2026  14:20             3,579 bandit_report_before.txt
27-10-2026  14:21               594 flake8_report_before.txt
27-10-2026  14:20             2,419 pylint_report_before.txt
               3 File(s)          6,592 bytes
               0 Dir(s)  244,399,206,400 bytes free
```

Updated Program

```
PS C:\Users\bjpsr\OneDrive\Desktop\L5_42_student_grade_system> .\venv\Scripts\Act
ivate.ps1
(venv) PS C:\Users\bjpsr\OneDrive\Desktop\L5_42_student_grade_system> python stud
ent_grade_system.py
Is S01 passing: True
Calculated GPA: 4.35
Student Hash: ac7438e86cb85ebd90f4d0abb235f9a28f112e1ec0b7995156f3d5e868d315f8
Registration PIN: 7868
Grade sheet exported to: grade_sheet.txt
(venv) PS C:\Users\bjpsr\OneDrive\Desktop\L5_42_student_grade_system>
```

After Reports

```
em *_report_*.txt
    Directory: C:\Users\bjpsr\OneDrive\Desktop\L5_42_student_grade_system


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----        27-10-2026     14:46           1996 bandit_report_after.txt
-a---l        27-10-2026     14:20           3579 bandit_report_before.txt
-a----        27-10-2026     14:46              0 flake8_report_after.txt
-a---l        27-10-2026     14:21            594 flake8_report_before.txt
-a----        27-10-2026     14:46            286 pylint_report_after.txt
-a---l        27-10-2026     14:20           2419 pylint_report_before.txt
```

```
PS C:\Users\bjpsr\OneDrive\Desktop\L5_42_student_grade_system> Get-Content pylint_report_after.txt

--------------------------------------------------------------------
Your code has been rated at 10.00/10 (previous run: 5.51/10, +4.49)

PS C:\Users\bjpsr\OneDrive\Desktop\L5_42_student_grade_system> Get-Content bandit_report_after.txt
python : [main] INFO    profile include tests: None
At line:1 char:1
+ python -m bandit -r student_grade_system.py > bandit_report_after.txt ...
+ ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
    + CategoryInfo          : NotSpecified: ([main]     INFO    profile include tests:
    None:String) [], RemoteException
    + FullyQualifiedErrorId : NativeCommandError

[main]  INFO    profile exclude tests: None
[main]  INFO    cli include tests: None
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.11.0
Run started:2026-10-27 09:16:35.733877+00:00

Test results:
        No issues identified.
```

```
PS C:\Users\bjpsr\OneDrive\Desktop\L5_42_student_grade_system> Get-Content bandit_report_after.txt

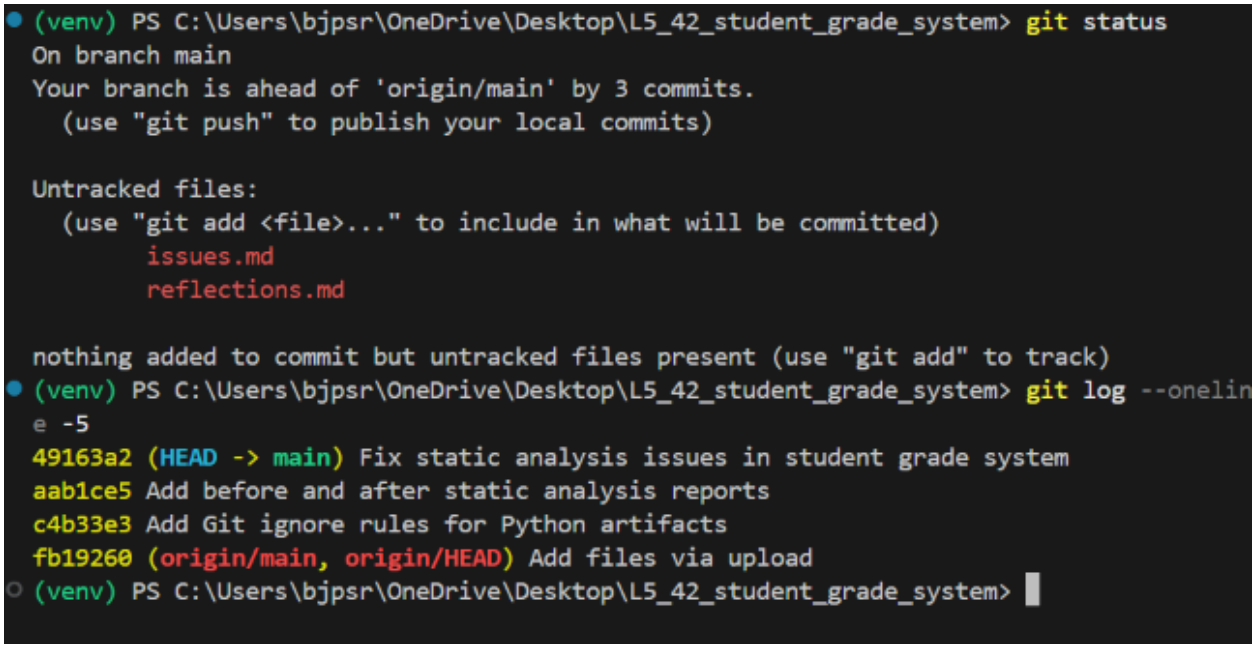
Test results:
        No issues identified.

Code scanned:
        Total lines of code: 68
        Total lines skipped (#nosec): 0
        Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
```

```
                High: 0
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
 Files skipped (0):
PS C:\Users\bjpsr\OneDrive\Desktop\L5_42_student_grade_system> Get-Content flake8_report_after.txt
PS C:\Users\bjpsr\OneDrive\Desktop\L5_42_student_grade_system>
```

Commits:


(venv) PS C:\Users\bjpsr\OneDrive\Desktop\L5_42_student_grade_system> git status
On branch main
Your branch is ahead of 'origin/main' by 3 commits.
(use "git push" to publish your local commits)
Untracked files:
(use "git add <file>..." to include in what will be committed)
issues.md
reflections.md
nothing added to commit but untracked files present (use "git add" to track)
(venv) PS C:\Users\bjpsr\OneDrive\Desktop\L5_42_student_grade_system> git log --onelin
e -5
49163a2 (HEAD -> main) Fix static analysis issues in student grade system
aab1ce5 Add before and after static analysis reports
c4b33e3 Add Git ignore rules for Python artifacts
fb19260 (origin/main, origin/HEAD) Add files via upload
(venv) PS C:\Users\bjpsr\OneDrive\Desktop\L5_42_student_grade_system>